MAZAK VQC-20/40

LinuxCNC / Mesa Retrofit — Complete Manual Set

Machine: Mazak Vertical Quality Center VQC-20/40 | **S/N:** 060231
Original control: Mazatrol M-2 (retired) | **New control:** LinuxCNC + Mesa 7i97T / 7i49 / 7i84U
Shop power: 460 VAC 3-phase (onboard transformer tapped for 460 V)
Document: MZK-LCNC-000 — Master Guide | **Revision:** A — July 2026

Volume 0 — Master Guide & Table of Contents

How This Manual Set Is Organized

This set documents the Mazak VQC-20/40 (SN 060231) **as converted** from its original Mazatrol M-2 control to LinuxCNC with Mesa Electronics FPGA hardware. It supplements — it does not replace — the original Mazak factory manuals, which remain authoritative for the base machine's mechanics, lubrication, and original electrical drawings.

Volume	Document No.	Title	Audience
0	MZK-LCNC-000	Master Guide & Table of Contents	Everyone
1	MZK-LCNC-001	Machine Overview & Safety	Everyone — **read first**
2	MZK-LCNC-002	Operation Manual (LinuxCNC)	Operators
3	MZK-LCNC-003	Mesa Conversion & Installation	Integrator / electrician
4	MZK-LCNC-004	Wiring Diagrams & I/O Reference	Integrator / electrician
5	MZK-LCNC-005	Maintenance & Troubleshooting	Maintenance
6	MZK-LCNC-006	Reference, Glossary & Index	Everyone

⚠ **PROJECT STATUS NOTICE.** As of this revision the retrofit is in the **planning / bring-up stage**. Most wiring assignments in Volume 4 carry a **PLAN_VERIFY** status: they are engineered plans that **must be traced and confirmed in the cabinet** before wires are landed or outputs energized. Statuses are printed with every pin table. Nothing in this set overrides the verification steps in Volume 3.

Master Table of Contents

Volume 1 — Machine Overview & Safety

Volume 6 — Reference, Glossary & Index

1. Original Mazak Manual Library (locations & applicability)
2. Key Project Files (repo & OneDrive paths)
3. External References
4. Glossary of Terms
5. Alphabetical Index (full set)

Source Documents

This manual set was compiled July 2026 from these authoritative project sources:

Source	Location
Original Mazak manuals (11 PDFs)	`OneDrive-Personal/OneDrive Docs/Mazak/Manuals_SN060231/`
Architecture decision	`GitHub/Github LinuxCNC/docs/architecture_decision.md`
Project status & TODO	`GitHub/Github LinuxCNC/docs/project_status.md`
Current pin authority (91 signals)	`GitHub/Github LinuxCNC/mesa/current_pin_authority.csv`
Wiring master workbook	`GitHub/Github LinuxCNC/wiring/Mazak_Wiring_Master_7i97T_7i49_7i84U.xlsx`
HAL / INI skeletons	`GitHub/Github LinuxCNC/linuxcnc/`
PLC signal dictionary (385 signals)	`Manuals_SN060231/02_Maintenance/YM2V39L_element_list.csv`

End of Volume 0.